

JAY MATHPAL

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Career Summary

Full-Stack Engineer building **scalable distributed systems** utilizing an **AI-first engineering approach** with expertise in **microservices, asynchronous processing, and fault-tolerant architectures**. Experienced in designing production-ready **RAG** workflows and high-performance relational databases. Proven algorithmic skills with **600+ DSA problems solved**.

Education

Birla Institute of Applied Sciences, Bhimtal

Bhimtal, Uttarakhand

B.Tech – Computer Science and Engineering (Aggregate: 77%)

2022 – 2026

Experience

Software Engineering Intern

Remote

GomindAI

2025

- Contributed to a NestJS platform serving 1000+ users within an **Agile framework**, participating in iterations and translating system features into scalable backend APIs.
- Implemented secure user authentication workflows and asynchronous API communication layers with cross-platform data synchronization.

Projects

NewsHub | Next.js, Node.js, Express.js, PostgreSQL, Redis, BullMQ, Elasticsearch, Docker

2025

- Decoupled heavy PDF processing from the core API tier using Redis and BullMQ, improving the API request-response cycle with a <1ms task enqueue time for non-blocking execution.
- Scaled background worker nodes to **5x concurrency**, establishing fault-tolerant queue execution and high horizontal processing throughput.
- Integrated **Elasticsearch for inverted-index full-text search**, delivering an optimized average query latency of **21ms** on benchmark data.

Self-Hosted LLM | React.js, Node.js, Python (FastAPI), Ollama, ChromaDB, PostgreSQL, Docker

2026

- Designed a microservices-based AI platform using Node.js and FastAPI LLM workers containerized with **Docker** for seamless environment isolation.
- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline utilizing text embeddings and localized vector search powered by **ChromaDB** to minimize LLM hallucinations during PDF data mining.
- Separated system architecture into a **precomputed heavy ingestion pipeline** and a **lightweight query pipeline** to decouple resource-intensive processing from user requests.

Finance Management Platform | Next.js, Node.js, Prisma, PostgreSQL, Clerk, Tailwind CSS

2025

- Designed a secure financial engine implementing data validation constraints and strict transactional safety logic on a relational **PostgreSQL** database.
- Integrated AI models for automated receipt scanning and structured data extraction to drive automated tracking workflows.
- Optimized query load times through database indexing strategies and cursor-driven pagination matrices. **Live:** finance-budget-platform-7uya.vercel.app

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL

Frontend: React.js, Next.js

Backend & Systems: Node.js, Express.js, FastAPI, REST APIs, Microservices, Asynchronous Processing

Databases & Queues: PostgreSQL, Redis, Elasticsearch, BullMQ, ChromaDB, Prisma

Tools & Infrastructure: Docker, Git, GitHub Actions, CI/CD pipelines

Coursework & Methodologies: Data Structures & Algorithms, Java OOPs, DBMS, Operating Systems, **Agile Framework**

Generative AI: Retrieval-Augmented Generation (RAG), Vector Embeddings, Semantic Search